NAMMATUTOR TOPPER NOTES

Solutions

— concentration, Raoult's law & colligative properties —

CLASS 12 · CHEMISTRY · CHAPTER 2

Types of Solutions

Concentration Terms

Henry's Law

Raoult's Law

Colligative Properties

Van't Hoff Factor

1 What is a Solution?

a homogeneous mixture of two or more components

DEFINITION

The component present in **larger quantity** is the **solvent**; every other component is a **solute**. Solutions can be solid-in-liquid, gas-in-liquid, solid-in-solid (alloys) — nine combinations in total, from the three states of matter.

✎ Four ways to measure "how much solute"

Term	Formula	Note
Mass %	(mass of solute ÷ mass of solution) × 100	Temperature-independent
Mole fraction (x)	moles of one component ÷ total moles	x(solute) + x(solvent) = 1
Molarity (M)	moles of solute ÷ litres of solution	Changes with temperature
Molality (m)	moles of solute ÷ kg of solvent	Temperature-independent — preferred for colligative properties

BOARD TRAP

Molarity uses the volume of the whole **solution**; molality uses the mass of only the **solvent**. This single difference is why molarity shifts with temperature (volume expands on heating) while molality does not — and it's exactly why colligative-property formulas are written in molality.

ONE-LINE RECALL

"Molarity moves with heat. Molality doesn't — because mass, unlike volume, never changes with temperature."

✎ Worked example — mass % to molality

TRY IT

Q: A solution is 10% NaCl by mass. Find its molality. (Molar mass NaCl = 58.5 g/mol)

Step 1: In 100 g of solution, NaCl = 10 g, water (solvent) = 90 g = 0.090 kg.

Step 2: Moles of NaCl = 10 ÷ 58.5 = 0.171 mol.

Step 3: Molality = moles of solute ÷ kg of solvent = 0.171 ÷ 0.090 = **1.90 m**

2 Henry's Law

how a gas dissolves in a liquid

THE LAW

$p = K_H \cdot x$ (partial pressure = Henry's constant × mole fraction of gas in solution)

Higher K_H → lower solubility

K_H rises with temperature

Used for: soda bottles, deep-sea diving (the "bends")

3 Raoult's Law

how a solute lowers the solvent's vapour pressure

THE LAW

$$p_{solution} = x_{solvent} \cdot p^{\circ}_{solvent}$$

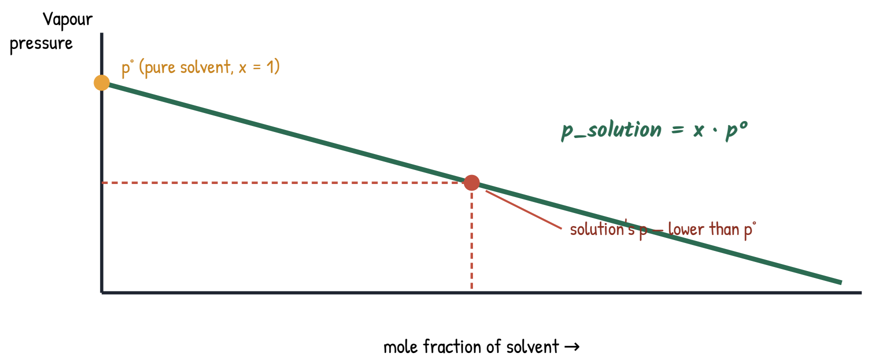

Fig 2.1 — vapour pressure falls in a straight line as solute is added

BOARD TRAP

Raoult's Law applies to **every component** in an ideal solution, not just the solvent. For an ideal mixture of two volatile liquids: $p_{total} = x_A p^{\circ}_A + x_B p^{\circ}_B$. Non-ideal solutions show positive or negative deviation — know both graphs.

✎ Ideal vs Non-ideal — three-line test

Type	ΔH_{mix}	Example
Ideal	0 — no heat change on mixing	Benzene + toluene
Positive deviation	> 0, endothermic; A–B forces weaker than A–A, B–B	Ethanol + acetone
Negative deviation	< 0, exothermic; A–B forces stronger than A–A, B–B	Chloroform + acetone

4 Colligative Properties

properties that depend only on the NUMBER of solute particles, not their identity

Property	Formula	What changes
Vapour pressure lowering	$\Delta p = x_{solute} \cdot p^{\circ}$	Falls
Boiling point elevation	$\Delta T_b = K_b \cdot m$	Rises
Freezing point depression	$\Delta T_f = K_f \cdot m$	Falls
Osmotic pressure	$\pi = MRT$	Solvent flows in

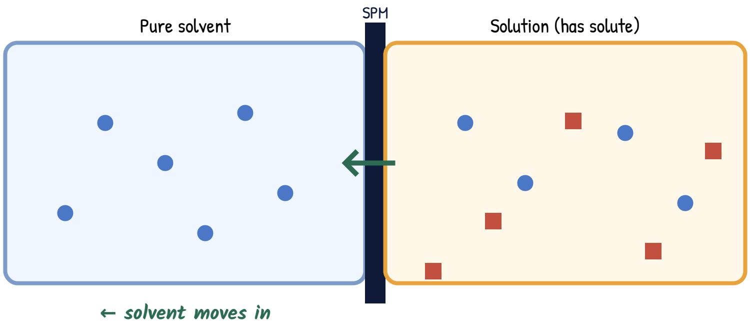

Fig 2.2 — osmosis: solvent crosses a semi-permeable membrane toward the solution, creating osmotic pressure π

WHY "COLLIGATIVE"

From Latin colligatus — "bound together". One mole of glucose and one mole of sucrose lower the freezing point by the **same amount**, even though the molecules are completely different — only the particle count matters.

✎ Constants worth memorising

Solvent	K_b (K kg/mol)	K_f (K kg/mol)
Water	0.52	1.86
Benzene	2.53	4.90
Acetic acid	3.07	3.90

5 Van't Hoff Factor (i)

the correction real solutes need

WHY WE NEED IT

Colligative-property formulas assume the solute stays as single, whole molecules. Electrolytes **dissociate** into more particles; some solutes **associate** into fewer. The Van't Hoff factor corrects for the mismatch.

Behaviour	Value of i	Example
Normal (no change)	$i = 1$	Glucose, urea
Dissociation	$i > 1$	$NaCl \rightarrow Na^+ + Cl^-$ ($i \approx 2$)
Association	$i < 1$	Ethanoic acid dimer in benzene ($i \approx 0.5$)

✎ Corrected formulas — just multiply by i

$\Delta T_f = i \cdot K_f \cdot m$ $\Delta T_b = i \cdot K_b \cdot m$ $\pi = i \cdot MRT$

BOARD TRAP

For NaCl, i is close to 2 but rarely exactly 2 — dissociation is never 100% complete in real solutions. Degree of dissociation **α** connects the two: $i = 1 + (n - 1)\alpha$, where n is the number of ions the compound splits into.

EXAM-LINE WORTH MEMORISING

"i is not a constant of the solute — it depends on concentration too, because dissociation is rarely complete."

✎ Worked example — finding i from ΔT_f

TRY IT

Q: A 0.1 m NaCl solution freezes at −0.352°C. K_f for water = 1.86 K kg/mol. Find i.

Step 1: $\Delta T_f = i \cdot K_f \cdot m$, so $i = \Delta T_f \div (K_f \cdot m)$

Step 2: $i = 0.352 \div (1.86 \times 0.1) = 0.352 \div 0.186 =$ **1.89**

Step 3: Close to 2 ($NaCl \rightarrow Na^+ + Cl^-$), confirming near-complete dissociation.